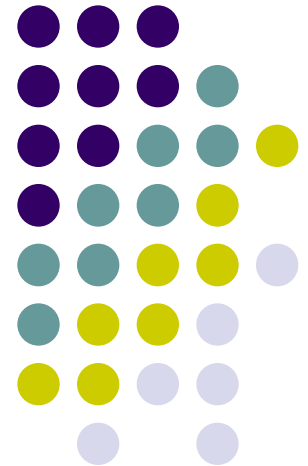
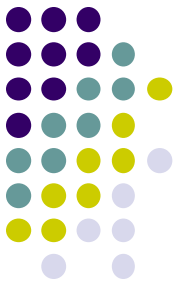


Dynamic Modelling for Human-centered Systems

Lecture 1

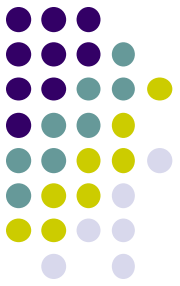
Reader: chapter 1 and 2





Agenda for today

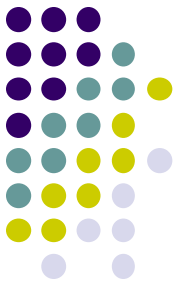
- What is a model?
- Why modelling?
- Designing models
- Growth model in Excel
- Assignment 1



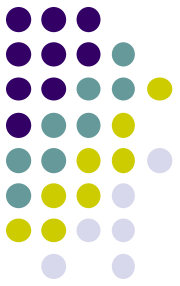
Agenda for today

- **What is a model?**
- Why modelling?
- Designing models
- Growth model in Excel
- Assignment 1

Model



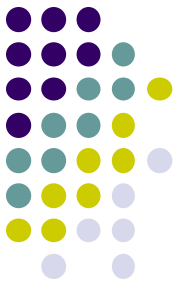
- What is a model?
 - Describe the most important characteristics of model
 - With a neighbour, answer via menti.com
- Definitions:
 - *A representation of an object, system, or idea in some form other than that of the entity itself (R.F. Shannon)*
 - *A simplified representation of a system or phenomenon, as in the sciences, with any hypotheses required to describe the system or explain the phenomenon, often mathematically (Random House Unabridged Dictionary)*
 - *A schematic description of a system, theory, or phenomenon that accounts for its known or inferred properties and may be used for further study of its characteristics (The American Heritage Dictionary)*



Types of models

- Knowledge driven
 - starting from human observations and interpretation
 - **knowledge engineering**
- Data driven
 - starting from data
 - **machine learning**

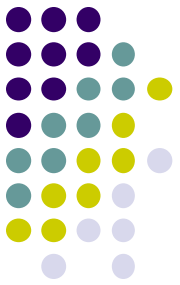
Example



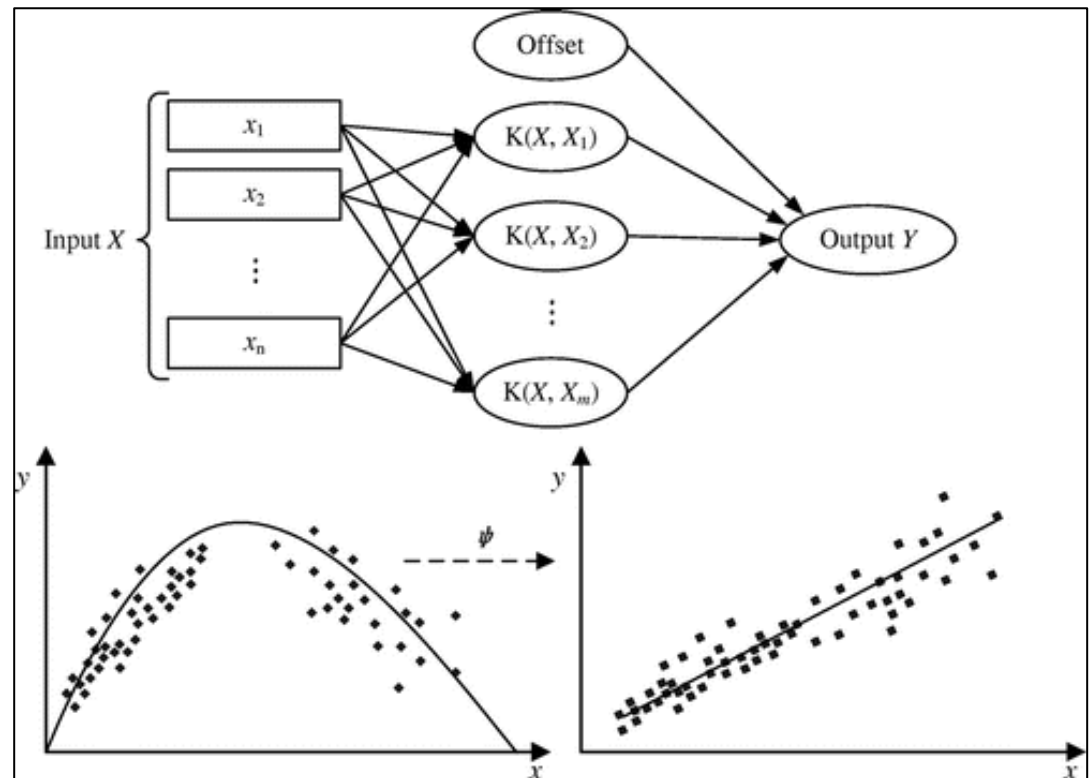
- Planetarium:
 - Representation of *planets and their orbit*
 - Aim:
 - Show working of solar system
 - ... and Eisinga: proof that the world will not disappear
 - Type
 - physical
 - *dynamic*



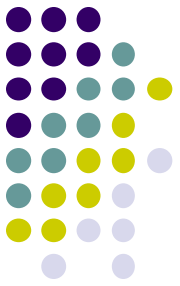
Another example



- Statistical / machine learning models
 - Representation of *correlation between input and output*
 - Aim:
 - predict output based on input
 - Type
 - numerical
 - *static*



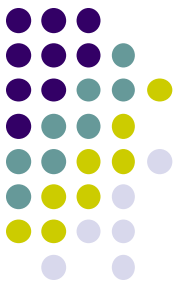
Yet another



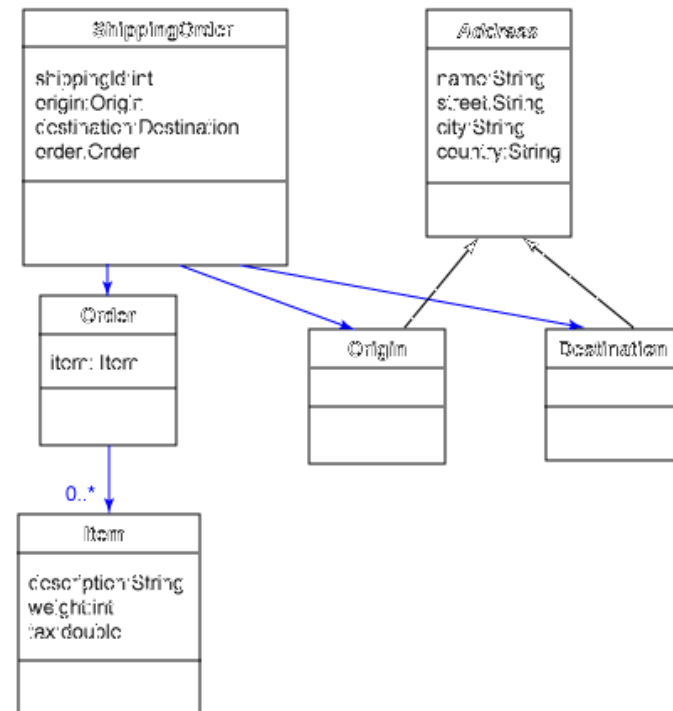
- Marquette (scale model):
 - Representation of *spatial properties of between objects*
 - Aim:
 - study spatial aspects beforehand
 - Type
 - physical
 - *static*



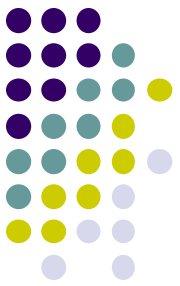
And again...



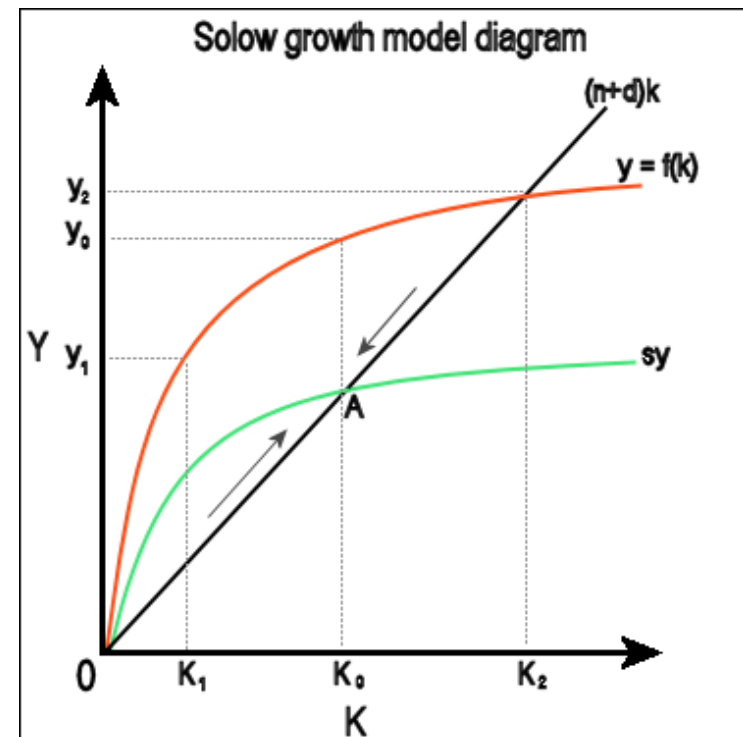
- Software model:
 - Representation of snippets of computer programs, their **functions and relations**
 - Aim:
 - verify program beforehand
 - simplify or automate programming
 - Type:
 - diagrams and semi-formal descriptions
 - **static (or dynamic)**

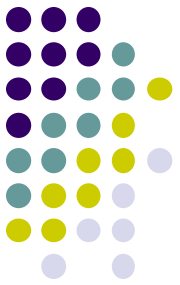


... another example



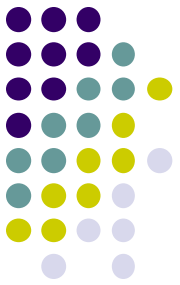
- Economic model:
 - Representation of **economic processes**
 - Aim:
 - predict aspects of economy
 - tool for determining economic policies
 - Type:
 - mathematical formulae
 - **dynamic**
- Many other types exist...





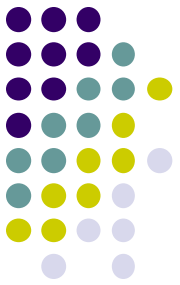
Our focus in this course

- We restrict ourselves to **dynamic models**
 - what is the most important property? Via menti.com
- A **dynamic model**:
 - describes a specific phenomenon, theory or system (i.e. a process)
 - specifies **changes over time**
 - i.e. can simulate the process



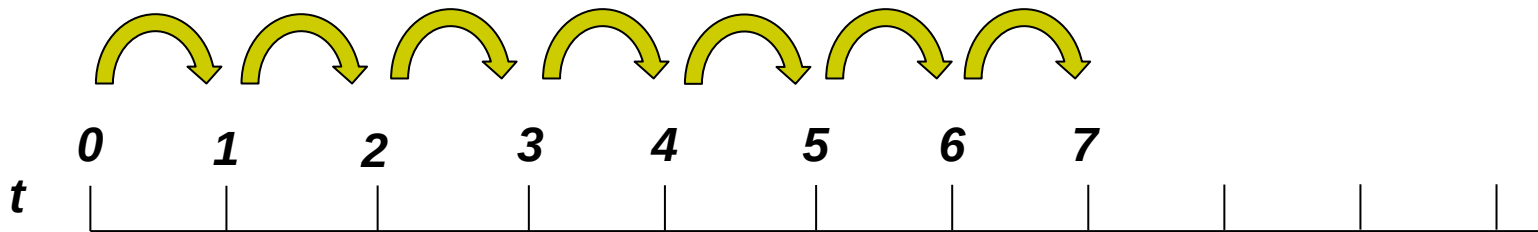
Risks of using models

- Which risks do you see risks in using models?
- Important observations
 - a model is a *simplification*
 - during the design of the models, *choices* have been made
 - assumptions
 - a model is not the same as reality!



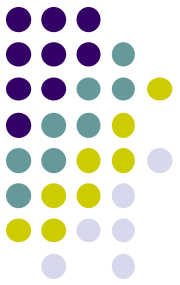
What is a dynamic model?

- Describes a **process** by defining how a next state is derived given the current state.



- At each point, only the current situation is considered, and the same rules are applied at every timepoint

Elements of a dynamic model

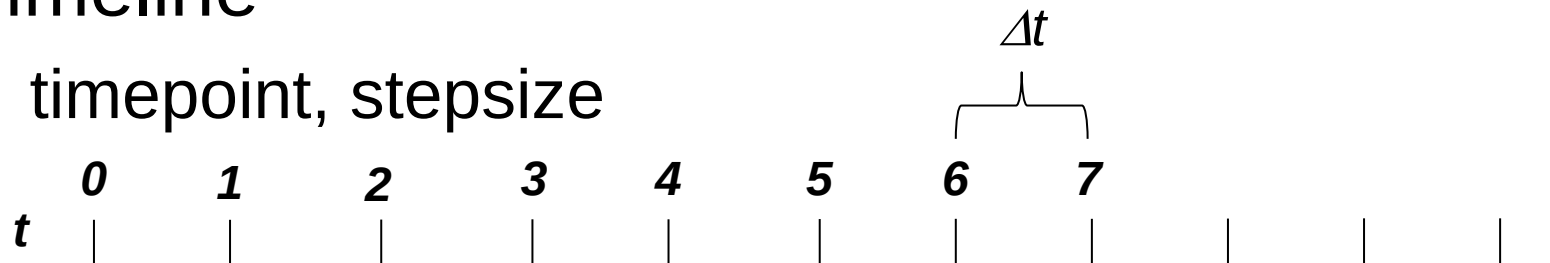


- Concepts

- represented as variables
 - temperature of an egg: *TempEgg*

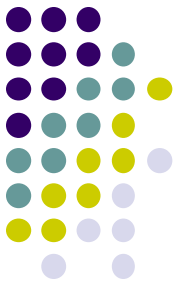
- Timeline

- timepoint, stepsize



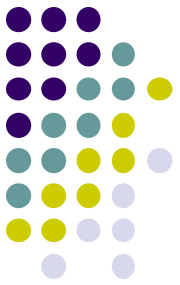
- Rules that describe how concepts change per time step

- $TempEgg(t + \Delta t) = \beta * TempWater(t) + (1 - \beta) * TempEgg(t)$



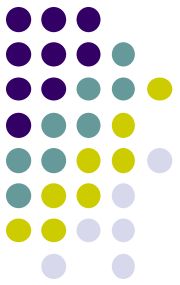
Additional elements

- Start values
 - $TempEgg(0) = 35$
- Optional parameters
 - $\beta = 0,33$
- Optional stochastic variables
 - random values



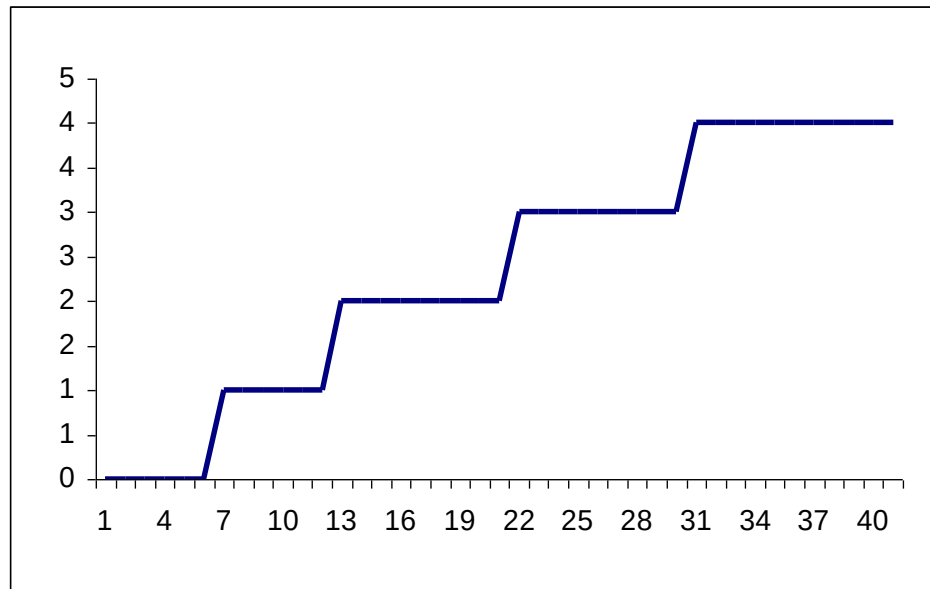
Model outcome

- How to describe the outcome of a *dynamic* model?
- List of states of a model at subsequent time points:
 - (simulation) trace
 - simulation run
- How can we show it?

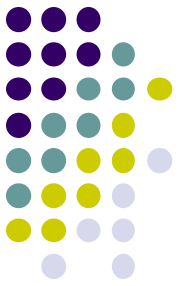


Visualisation of simulations

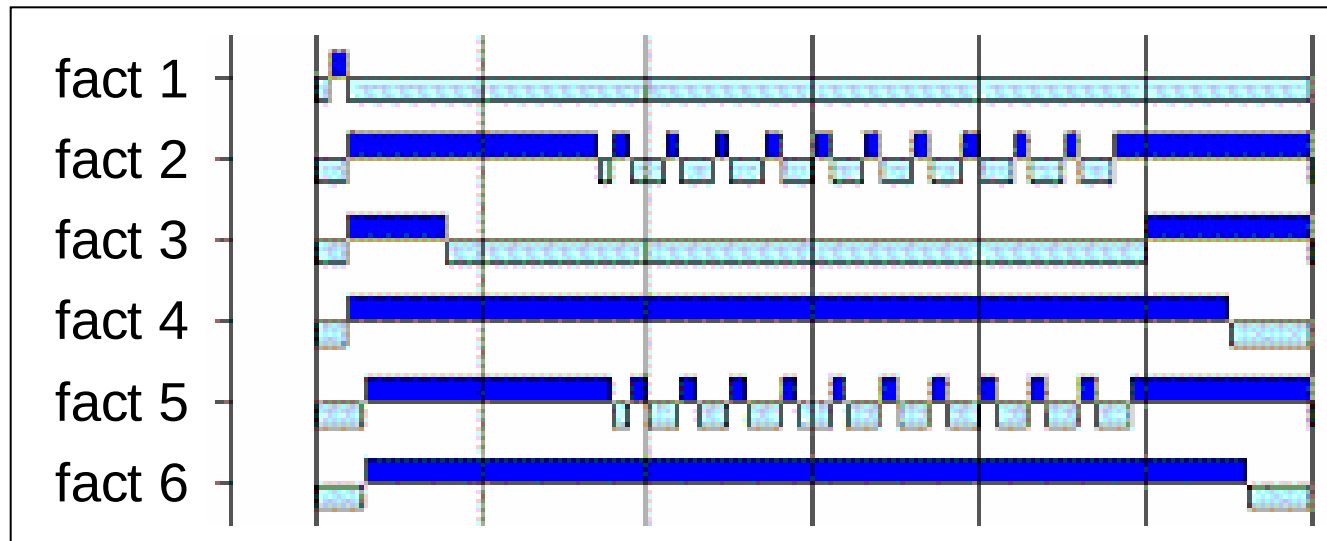
- Dependent on the representation of the model
 - graphical representation of values of model variables at subsequent time points

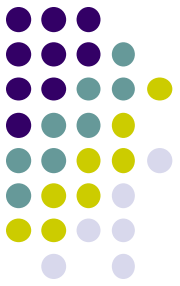


Visualisation - 2



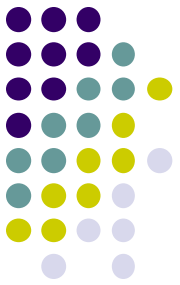
- representation of truth values of model variables at subsequent time points (state trace)





Agenda for today

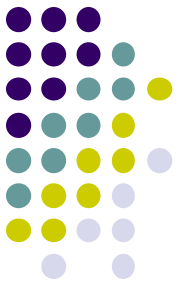
- What is a model?
- **Why modelling?**
- Designing models
- Growth model in Excel



Why modelling?

1. Study a process to increase understanding:
 - study the behaviour of a process over time
 - performing “what if” simulations
2. Basis for intelligent systems
 - models are used to understand what is happening with humans (and what to do)
 - models can make systems behave human-like

Users of models



- People:

- researchers
- policy makers



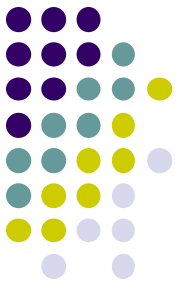
- Systems:

- adaptive systems
- human-like systems

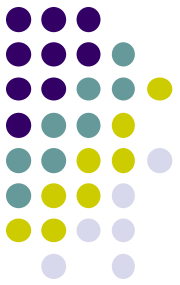
⇒ *second half of the course*



Use of models for intelligent applications

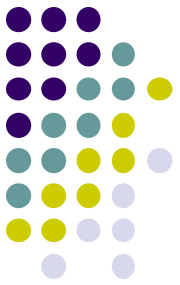


- Ambient Intelligence:
 - Intelligent systems support humans in their daily environment
- Requirements:
 - Systems have *knowledge* about human functioning
 - This course: making dynamic models about human functioning and incorporate them in systems



Agenda for today

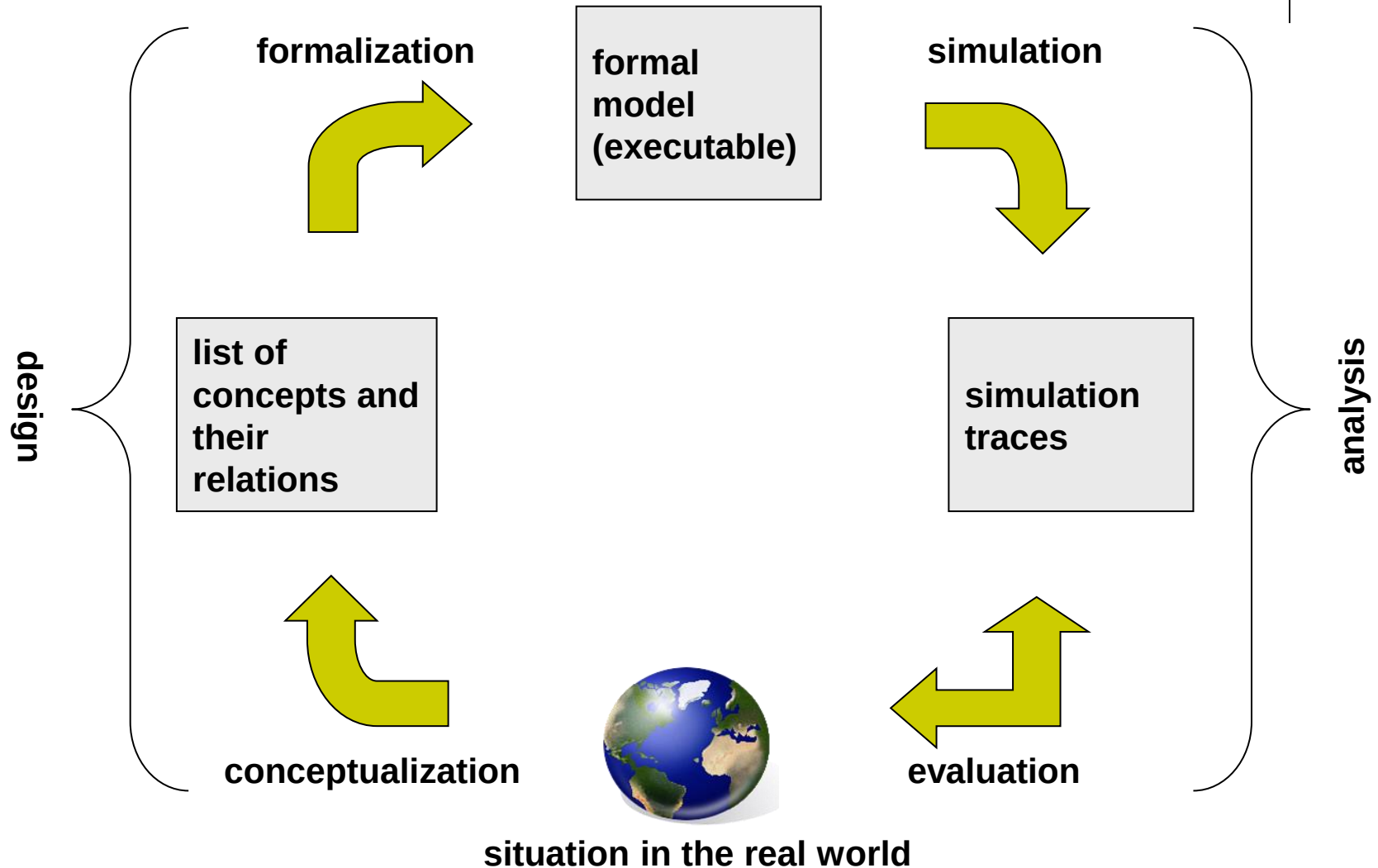
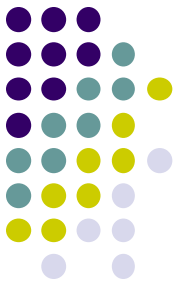
- What is a model?
- Why modelling?
- **Designing models**
- Growth model in Excel



Designing models

- Quotes:
 - *"Make your theory as simple as possible, but no simpler."* (Albert Einstein)
 - *"For every complex question there is a simple and wrong solution."* (H.L. Mencken)
- Creative process!
 - requires choices and assumptions
 - not one correct solution
 - global / detailed; what to include / ignore
- Guideline provides structure to process

Modelling and Simulation Cycle





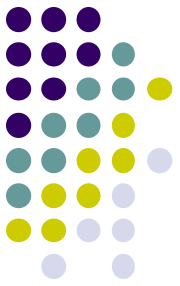
1. Conceptualisation

- Several steps
- Iterative process
 - outcome task can influence other choices

A. Characterizing the process

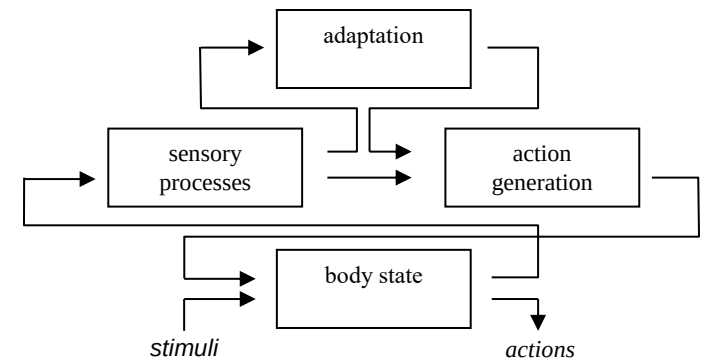
- scope and boundaries
- type of questions to answer with the model
- expected behaviour (***characteristic patterns***)

Conceptualisation (2)



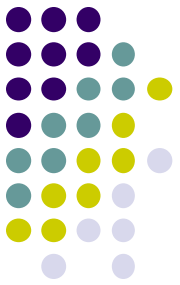
B. If necessary: describe sub processes

- identify sub processes
- determine order
- result: simple diagram with processes and arrows



C. Identify important concepts

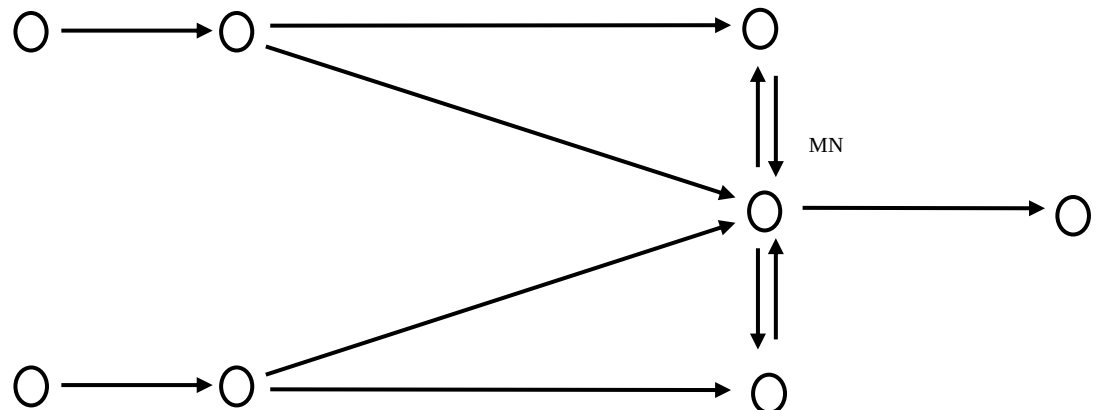
- which *factors* play a role in the process
 - objects, events, principles
- result: list of concepts

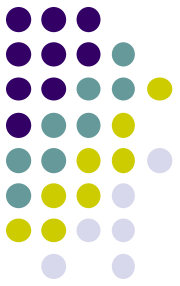


Conceptualisation (3)

D. Relations between concepts

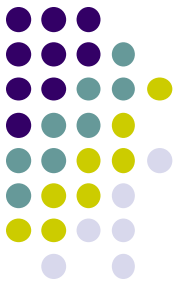
- which concepts affect each other
 - whether, not how
- most relations occur within a sub process
- result:
 - list “A affect B”
 - diagram





2. Formalisation

- Specification of the details of the conceptualization
 - concepts & relations
- Dependent on chosen representation
- Two main variant:
 - numerical representation
 - logical representation



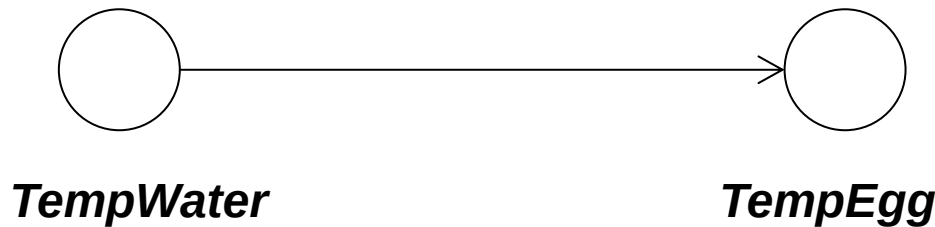
Numerical representation

- Concepts:
 - variables met numerical value
 - integer, real, 0 or 1
 - example: concept “temperature” → “Temp: real”
- Relations:
 - calculation rules to determine value of concept at next time point based on values other concepts
 - in dynamic model values change over time
 - Δt (change in time) is time step between two states

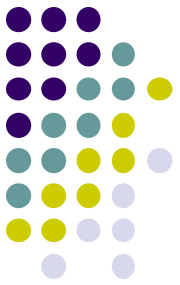


Numerical representation (2)

- Suppose: model of the temperature of an egg that is boiled in a pot of water

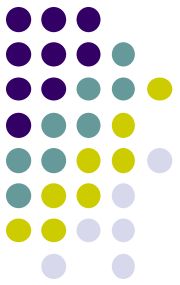


- Example calculation rules:
 - $TempEgg(t + \Delta t) = TempWater(t)$
 - $TempEgg(t + \Delta t) = \beta * TempWater(t) + (1 - \beta) * TempEgg(t)$



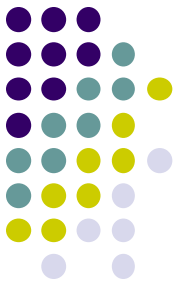
Logical representation

- Concepts:
 - statements (propositions) that can be *true* or *false*
 - example: `medicine_taken(hiv_slower)`
- Relations:
 - Boolean rules that specify which combination of truth values is required for a concept to be true at next time point



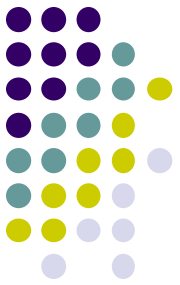
Logical representation (2)

- Example rules:
 - $A \wedge B \rightarrow C$:
 - if A and B are true, C becomes true
 - $A \vee B \rightarrow C$:
 - If A or B or both are true, C becomes true
 - $A(X) \rightarrow B(X)$:
 - If A is true for argument value X, B becomes true for the same argument X
 - $\text{not}(A) \rightarrow B$:
 - If A is not true, B becomes true
 - $A(X) \wedge X > 52 \rightarrow B$:
 - If A is true for some value X larger than 52, B becomes true



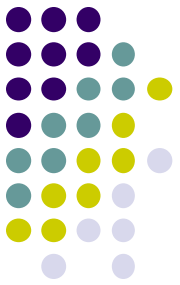
3. Simulation

- Expected behaviour of correct model is specified as characteristic pattern $\Rightarrow$ property
 - already during conceptualisation
- Define different scenarios that should be able to show properties
 - different start values
 - parameters that are different
 - stochastic variables (random factors)



4. Evaluation

- Simulations results are compared to expectations
 - hypotheses about model behaviour
- Used to draw conclusions about correctness of model
 - validation

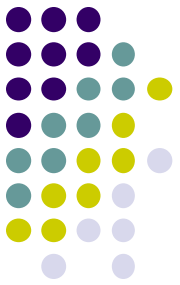


New term: model property

- A hypothesis about a model behavior (i.e. the outcome of a simulation given some input and parameter value) is called:

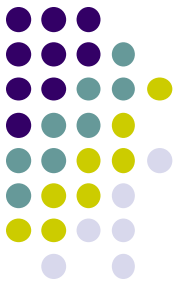
a model property

- When formulated in the design phase, it is also called “characteristic pattern” of a model



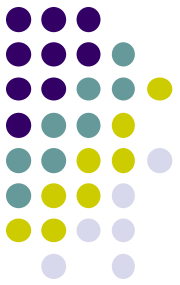
Model properties

- Can be statements about
 - effects that happen over time
 - combination of inputs that lead to some output
- Examples:
 - *Jupiter and Pluto will not collide within two years*
 - *when taking a pill every 3 hours, the medication level is between 12 and 20 mg/l*
- What is essential for a property?
 - it can be tested in an unambiguous way



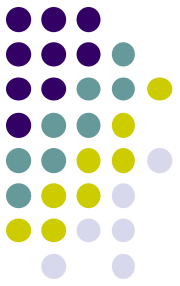
Validation of properties

- Run (several) simulations of the model and check the properties $\Rightarrow$ scenarios
 - choose conditions / input parameters intelligently
 - be precise about what you can conclude
- Alternatively:
 - use logic / mathematics to (dis)prove the properties
 - usually quite complex



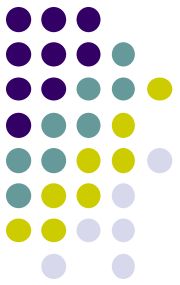
Steps in brief

1. **Conceptualisation:** identification of important concepts and relations
2. **Formalisation:** describe concepts and relations in an unambiguous way
3. **Simulation:** running experiments with models to generate simulations traces
4. **Evaluation:** verify whether the simulations are in line with expectations (properties about characteristics patterns)



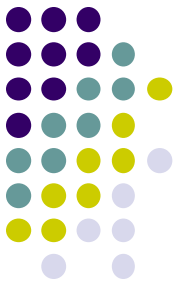
Wrap up:

- What is a property of model?
- Which two use cases have we seen?



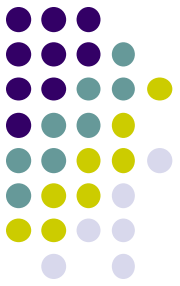
Reporting

- Structured reporting is important
 - many arbitrary choices in process
- Elements of report:
 - assumptions, choices, motivation
 - concepts, relations, expected patterns
 - scenarios used in simulation experiments
 - settings
 - input parameters
 - Verified properties, conclusions



Summary

- Dynamic models describe process in which factors change over time
- Models consists of:
 - concepts
 - rules
- Hypothesis / properties can be formulated and validated against simulation
 - for evaluating expected patterns
 - for studying the process



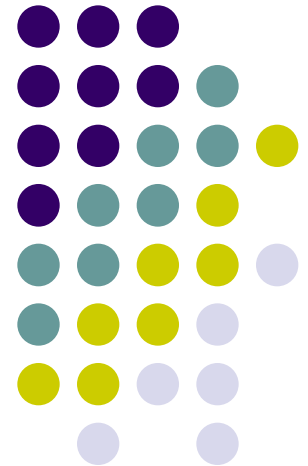
Agenda for today

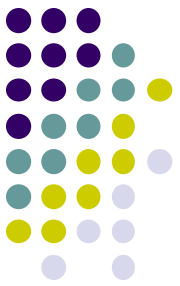
- What is a model?
- Why modelling?
- Designing models
- **Growth model in Excel**

Dynamic modelling for Human-centered Systems

Modelling growth

Chapter 2

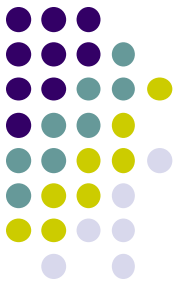




Modelling “growth”

- Basic example of model of dynamic process
- At the same time: building block for models of other process in which growth plays a role
- Aim: generic model
- Inspiration from nature

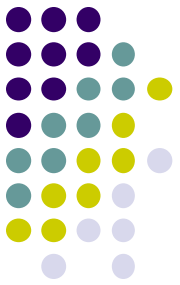




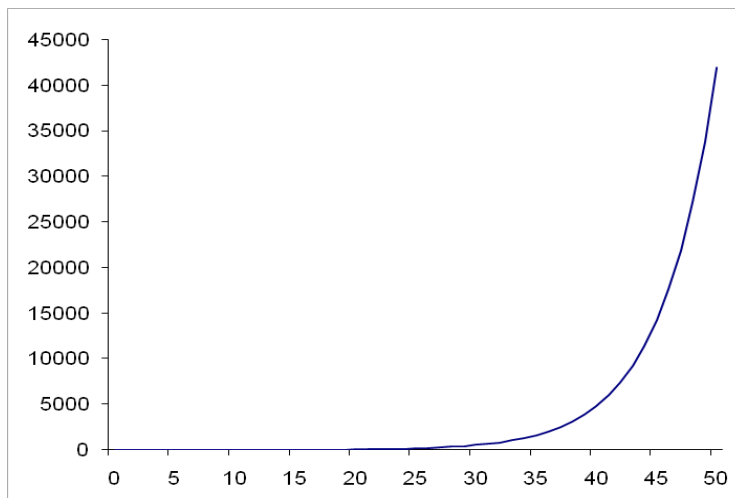
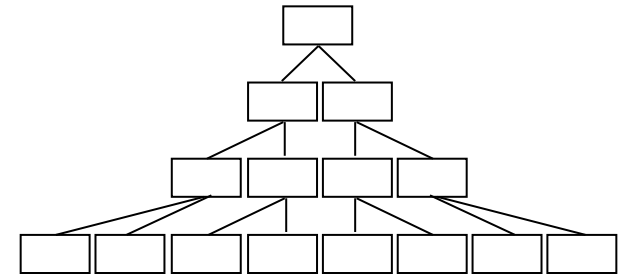
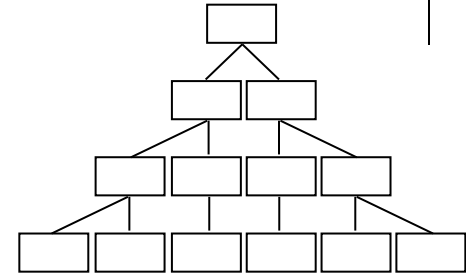
Understanding growth

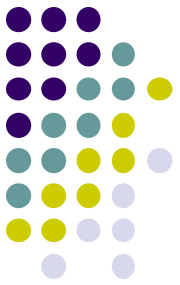
- Phenomenon growth:
 - generic process
 - economy, biology
 - relations, trust
- Inspiration: biological growth
 - two causes of growth at cell level
 - volume increase
 - cell division
 - two types of growth via cell division
 - every cell divides
 - only cells at the end divide

Generic growth patterns



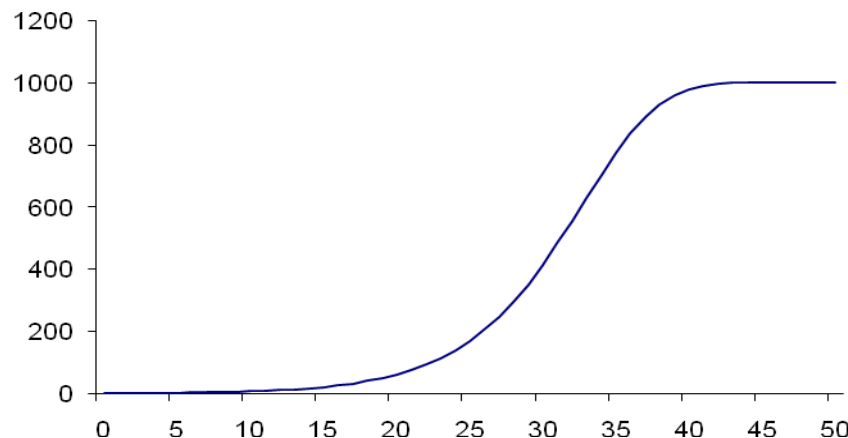
- Linear growth
 - division by fixed number of cells
 - equal gain per time step
- Exponential growth
 - division by every cell
 - every time step doubling

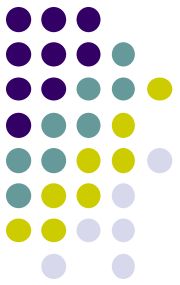




Generic growth patterns (2)

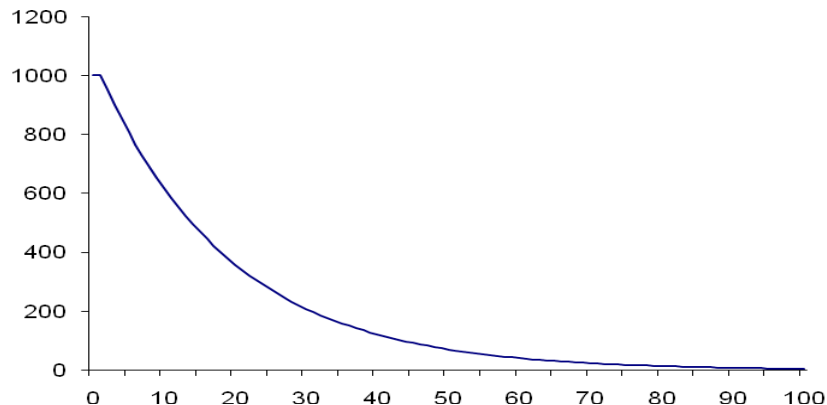
- Limited growth
 - growth decreases when some maximum is approached
 - growth is linear or exponential at start, then decreasing gain per time step

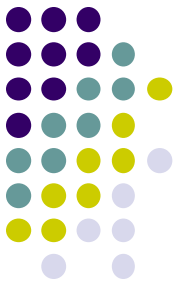




Generic growth patterns (3)

- Negative growth
 - loss is larger than the gain
 - exists in several variants
 - linear
 - exponential...

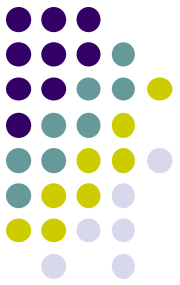




Which concepts?

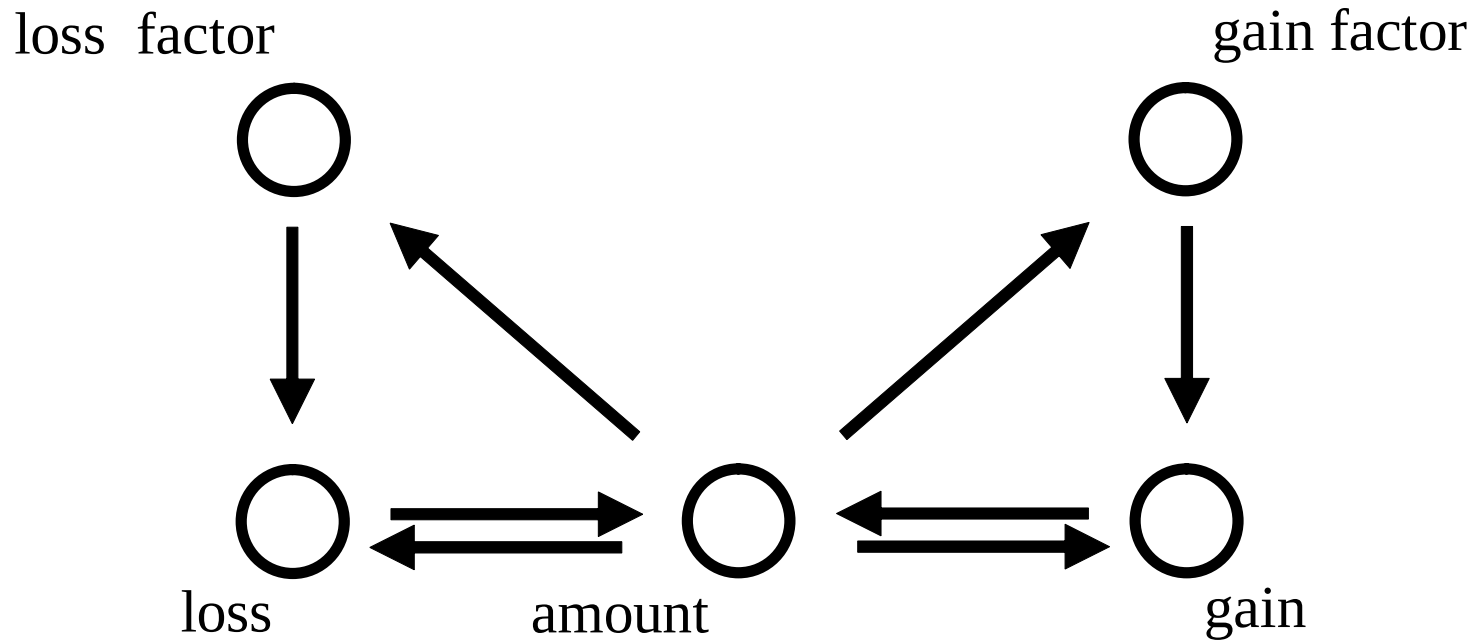
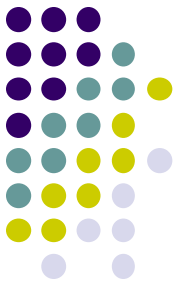
- What would be useful concepts for a dynamic model of growth?
 - answer via [menti.com](https://www.menti.com)

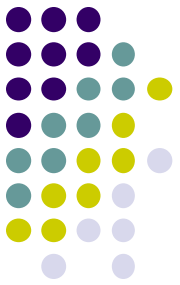
Conceptualisation of basic version



- Important concepts
 - amount
 - gain per time step
 - loss per time step
 - gain factor = gain as proportion of the total amount
 - loss factor = loss as proportion of the total amount

Visual representation of model





Formalisation

- **Amount and gain factor affect gain**

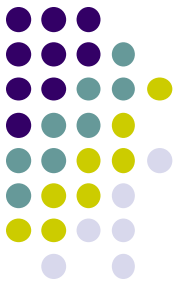
Gain is calculated by multiplying the gain factor by the amount:

$$Gain(t) = GainFactor(t) * Amount(t)$$

- **Amount and loss factor affect loss**

Loss is calculated by multiplying the loss factor by the amount:

$$Loss(t) = LossFactor(t) * Amount(t)$$



Formalisation - 2

- **From gain and loss to amount**

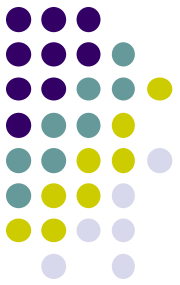
Gain adds to the amount and loss subtracts from the amount, both taken over the time interval that is passing.

$$Amount(t+\Delta t) = Amount(t) + (Gain(t) - Loss(t)) * \Delta t$$

- **Gain factor**

Constant *factor* for exponential growth

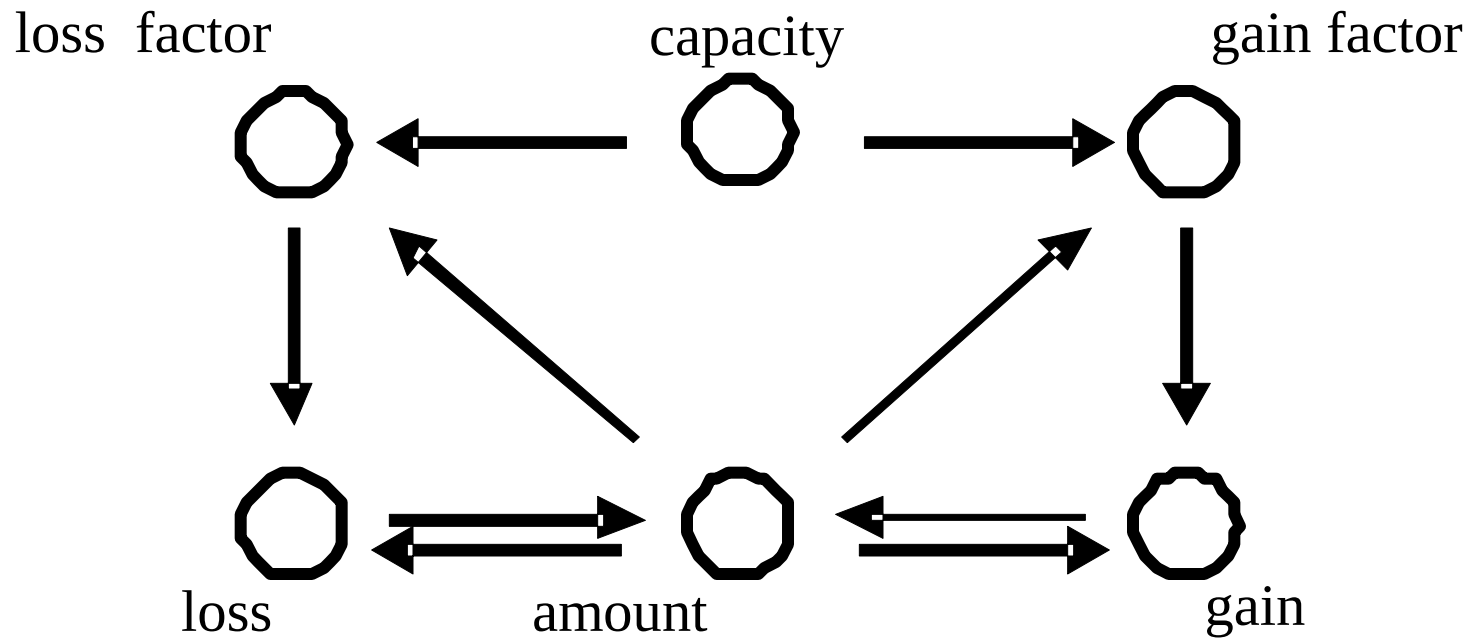
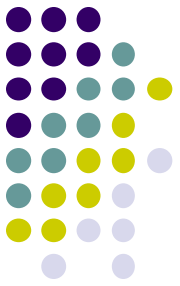
$$GainFactor(t) = \alpha$$

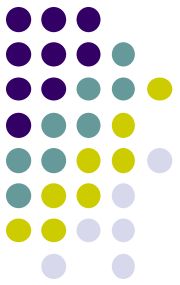


Limitation

- Model only allows linear and exponential growth
- Limited growth can not be represented
- Which change is needed?

Limited growth model

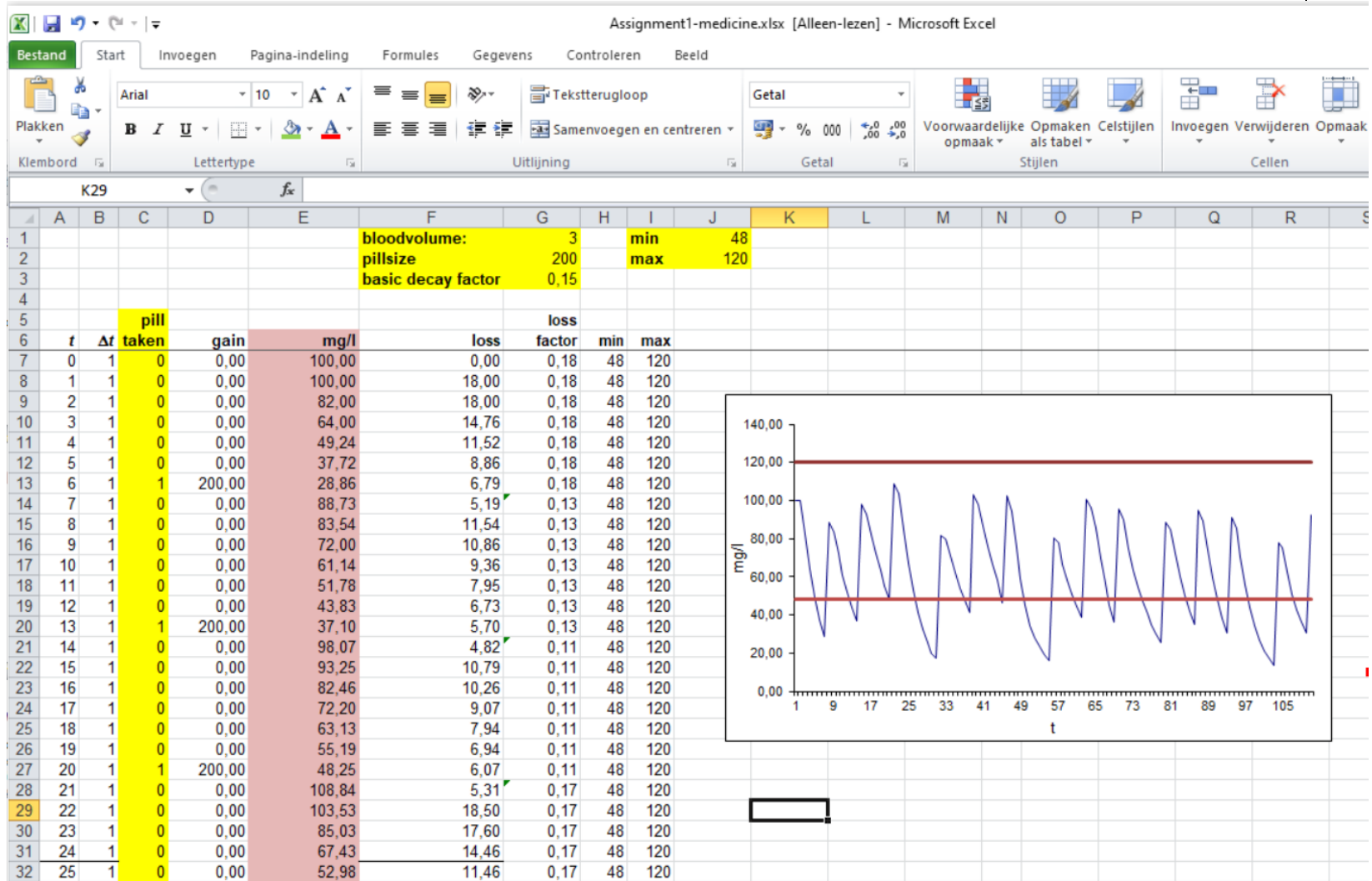
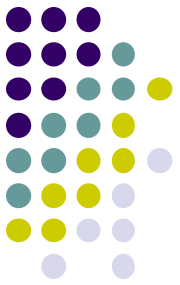




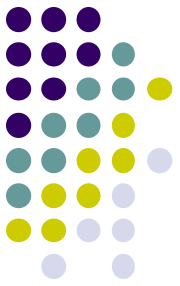
Adapted formalisation

- Limited *gain*:
 - proportional to distance to maximum capacity (C)
 - when close to C, the gain is also small
$$(C - Amount(t)) / C$$
 - thus:
 - $GainFactor(t) = \alpha * ((C - Amount(t)) / C)$
 - results in “logistic” growth

Modelling in Excel

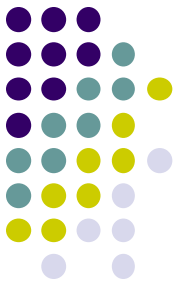


How to make dynamic models in Excel?



- Excel can be used to represent dynamic models
 - concepts → columns
 - timeline → subsequent rows
 - calculation rules → formulae in columns of cells
 - parameters → some specific cells

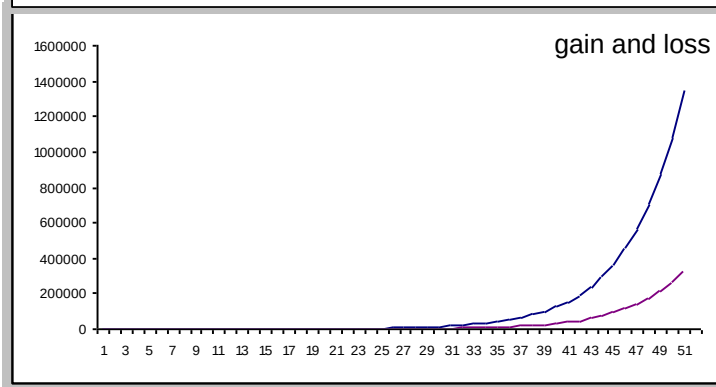
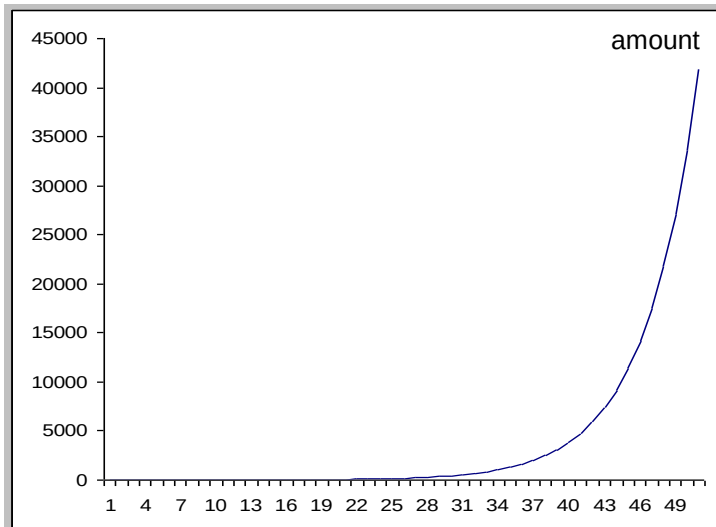
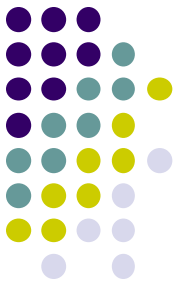
Example



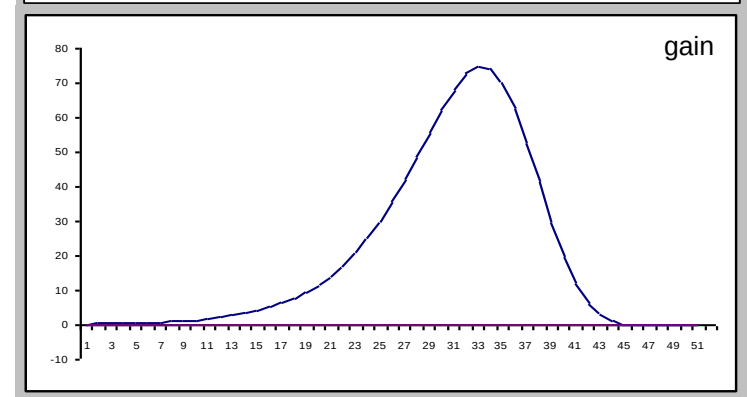
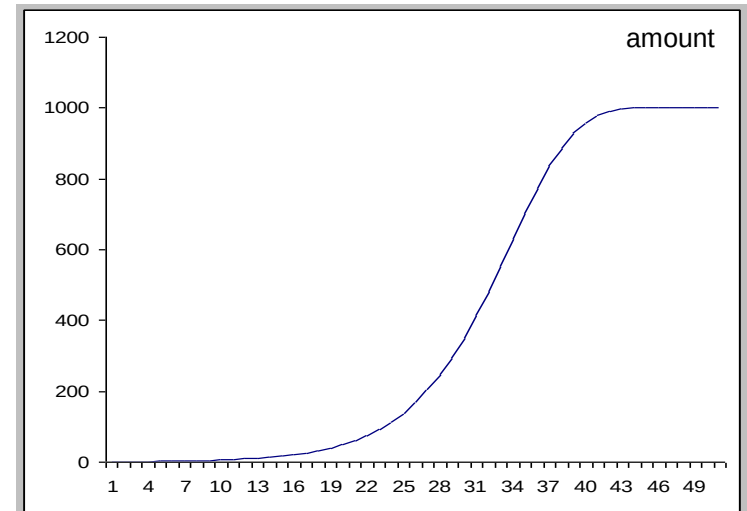
t	loss factor	loss	amount	gain	gain factor
0	0.1	10	100	40	0.4
1	0.1		130		0.4
2	0.1				0.4
3	0.1				0.4
4	0.1				0.4

The diagram illustrates the calculation of the amount at time t=1. It shows a sequence of values over time t from 0 to 4. The columns are: t, loss factor, loss, amount, gain, and gain factor. At t=0, the loss factor is 0.1, the loss is 10, the amount is 100, the gain is 40, and the gain factor is 0.4. At t=1, the loss factor is 0.1, the amount is 130, and the gain factor is 0.4. Arrows indicate the calculation of the amount at t=1: the loss factor (0.1) at t=0 points to the loss (10) at t=0, which then points to the amount (130) at t=1. The amount (100) at t=0 points directly to the amount (130) at t=1. The gain (40) at t=0 points to the amount (130) at t=1. The gain factor (0.4) at t=0 points to the gain factor (0.4) at t=1.

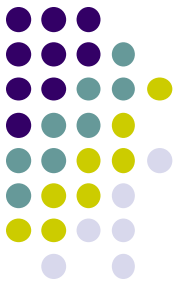
Results simulations



exponential

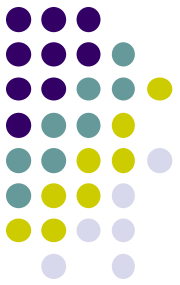


logistic



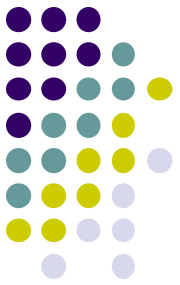
Assignment 1

- Understand specific model in Excel
- Make changes
- Formulate hypothesis about simulation result
- Validate hypothesis by comparing simulation outcomes



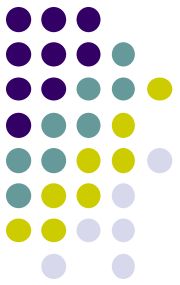
Model for assignment

- Prediction of drug level in blood
 - when taking medication, it is important to keep the drug level in the blood between certain limits
- Model
 - based on growth model
 - time steps of one hour
 - taking a pill increases medication level in blood
 - medication is slowly removed from body (decay factor)
 - decay factor depends on stochastic variable (mimics “unpredictable” physical activity)



What are good hypothesis?

- It can be tested!
 - unambiguous & precise
- What do you think?
 - sit with three neighbours
 - formulate one
 - submit some examples via [menti.com](https://www.menti.com)
- Examples:
 - the amount is always below 600 mg/l
 - when taking medication, the gain is larger than the loss
 - for a given activity level, the drug level never goes below 200 mg/l



Instructions for assignment

- Hand in a self-contained report
 - explain & illustrate
 - sketch context, discuss consequences
- Subtraction of points if the following is not done:
 - unambiguousness in the formulation of hypotheses;
 - precise, complete and logically correct description of the interpretation of the simulation results;